Active and passive random walks on stochastic temporal networks

Leo SPEIDEL[1,2], Renaud LAMBIOTTE[3], Kazuyuki AIHARA[1,4], and Naoki MASUDA[5,*]

[1]Department of Mathematical Informatics, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113-8656 Japan
[2]JST, ERATO, Kawarabayashi Large Graph Project, 2-1-2 Hitotsubashi, Chiyoda-ku, Tokyo, 101-8430 Japan
[3]Department of Mathematics/Naxys, University of Namur, rue de Bruxelles 61, B-5000 Namur, Belgium
[4]Institute of Industrial Science, The University of Tokyo, 4-6-1 Komaba, Meguro-ku, Tokyo, 153-8505 Japan
[5]Department of Engineering Mathematics, University of Bristol, Merchant Venturers Building, Woodland Road, Bristol, BS8 1UB UK

E-mail: *naoki.masuda@bristol.ac.uk

Abstract

Effects of the interevent time distribution on diffusion processes still remain poorly understood. So far, studies mainly adapted a data-driven or Markovian approach. In contrast, we derive analytical solutions for the steady state and mean recurrence time for random walks on stochastic temporal networks, where interevent time distributions are controlled explicitly.

Keyword: Random walks, Temporal networks, Interevent time distribution

1 Introduction

An accumulating amount of research has evinced non-trivial temporal patterns in networked systems. For example, the distribution of the interevent time ubiquitously follows a power law [4]. We analyse two versions of random walks on a stochastic temporal network model for which the interevent time distribution is controlled explicitly for every link [1]. This is in contrast to previous approaches, which were mainly data-driven or adapted a Markovian perspective [2].

2 Model

In the active random walk, interevent times of all links are assumed to be renewed after each transition. In contrast, in the passive random walk, interevent times are not renewed on all but one link, and transition rates depend on the entire path of the random walker. For the latter, we derived an approximation scheme that induced a type of correlated random walk.

3 Results

So far [3], we found that the steady state and mean recurrence time are (in)dependent of the interevent time distribution for the active (passive) random walk. We also confirmed a high accuracy of the approximations in numerical simulations. Further simulations suggest that similar results are obtained in the non-ergodic phase, where a random walker needs an infinitely long time on average to hop to an adjacent node.

Acknowledgements

N.M. acknowledges JST, CREST, and JST, ERATO, Kawarabayashi Large Graph Project.

References

[1] T. Hoffmann, M. A. Porter, and R. Lambiotte, "Generalized master equations for non-Poisson dynamics on networks", *Phys. Rev. E*, vol.86, 046102 (2012).

[2] P. Holme and J. Saramäki, "Temporal networks", *Phys. Rep.*, vol.519, 97–125 (2012).

[3] L. Speidel, R. Lambiotte, K. Aihara, and N. Masuda, "Steady state and mean recurrence time for random walks on stochastic temporal networks", *arXiv:1407.4582* (2014).

[4] A. Vázquez, J. G. Oliveira, Z. Dezsö, K.-I. Goh, I. Kondor, and A.-L. Barabási, "Modeling bursts and heavy tails in human dynamics", *Phys. Rev. E*, vol.73, 036127 (2006).